CodeAlpha Robotics & Automation Internship — Task 2

Design, Kinematic Modeling, CAD Assembly, and Motion Simulation of a 4-DOF Robotic Manipulator

Author: CodeAlpha Robotics Intern

Domain: Robotic Arm Kinematics & Simulation

Deliverables: 5 CAD STLs, 1 OBJ Assembly, 3D Engine, Offline WebGL

Unit Convention: Metric (Lengths in cm, Angles in °)

1. Executive Overview & Technical Objectives

Articulated robotic manipulators form the backbone of modern flexible manufacturing and pick-and-place intralogistics. This engineering report documents the kinematic design, closed-form analytical solvers, solid CAD component modeling, and physical trajectory verification of a 4-DOF articulated manipulator equipped with a parallel-jaw end-effector. All geometric lengths are defined in metric centimeters (cm), angles in degrees (°), and positional accuracy tolerances are held within 1.0 mm.

2. Specification of Degrees of Freedom (DOF)

The manipulator incorporates **4 active revolute degrees of freedom (4-DOF) plus a 1-DOF prismatic dual-finger pneumatic gripper**. This configuration provides spatial positioning (X, Y, Z) and pitch attitude regulation, optimized for pick-and-place material transfer.

Joint	Type	Axis of Motion	Angular Range	Link Length	Functional Role
Joint 1 (Base)	Revolute	Yaw (Z-axis)	-180° to +180°	L1 = 15.0 cm	Workcell azimuthal sweep
Joint 2 (Shoulder)	Revolute	Pitch (Y-axis)	-30° to +110°	L2 = 25.0 cm	Primary vertical and radial reach
Joint 3 (Elbow)	Revolute	Pitch (Y-axis)	-120° to +120°	L3 = 20.0 cm	Forearm altitude and extension
Joint 4 (Wrist)	Revolute	Pitch (Y-axis)	-90° to +90°	L4 = 10.0 cm	Tool Center Point (TCP) alignment
End-Effector	Prismatic	Linear Pinch	0 to 4.0 cm stroke	Span: 5.0 cm	Positive grasp of target payloads

3. Kinematic Modeling & Denavit-Hartenberg (DH) Parameters

Coordinate frames are attached at each joint axis according to standard Denavit-Hartenberg convention. The DH kinematic parameters governing homogeneous link transformations are detailed in Table 2:

Link (i)	Link Length (a_i)	Link Twist (α_i)	Link Offset (d_i)	Joint Variable (θ_i)
Link 1	0.0 cm	+90°	15.0 cm (L1)	θ_1 (Variable)
Link 2	25.0 cm (L2)	0°	0.0 cm	θ_2 (Variable)
Link 3	20.0 cm (L3)	0°	0.0 cm	θ_3 (Variable)
Link 4	10.0 cm (L4)	0°	0.0 cm	θ_4 (Variable)

Analytical Inverse Kinematics Formulation: For target Tool Center Point position (X_{tgt}, Y_{tgt}, Z_{tgt}) and desired tool approach angle ψ, the base azimuth is resolved as $\theta_1 = \text{atan2}(Y_{tgt}, X_{tgt})$. The wrist center coordinates are isolated via $r_w = \text{sqrt}(X_{tgt}^2 + Y_{tgt}^2) - L4 \cdot \cos(\psi)$ and $z_w = Z_{tgt} - L4 \cdot \sin(\psi)$. Applying the Law of Cosines to the shoulder-elbow-wrist triangle yields $\cos(\theta_3) = (r_w^2 + (z_w - L1)^2 - L2^2 - L3^2) / (2 \cdot L2 \cdot L3)$, providing closed-form, deterministic solutions with elbow-up configuration without iterative solver latency.

4. Kinematics Verification & Error Analysis

To rigorously demonstrate that the kinematics formulation is analytically validated, the verification script (verify_kinematics.py) executes Inverse Kinematics across all primary workcell waypoints, substitutes the solved joint angles back into Forward Kinematics, and computes Cartesian Euclidean position error. Table 3 presents the quantitative results, verifying **max error < 0.001 mm** and zero joint limit violations across the full trajectory.

Waypoint Name	Target (X, Y, Z) [cm]	Solved Joint Angles (θ_1, θ_2, θ_3, θ_4) [°]	Position Error	Validation Status
Pick Hover	(24.0, 16.0, 20.0)	(33.7°, 71.9°, -99.9°, -42.1°)	0.0000 mm	PASSED (In Limits)
Pick Grasp Target	(24.0, 16.0, 4.0)	(33.7°, 42.5°, -106.8°, -15.7°)	0.0000 mm	PASSED (In Limits)
Safe Transit Lift	(24.0, 16.0, 26.0)	(33.7°, 79.7°, -94.0°, -45.7°)	0.0000 mm	PASSED (In Limits)
Place Swing Hover	(-18.0, 25.0, 26.0)	(125.8°, 75.3°, -88.4°, -46.9°)	0.0000 mm	PASSED (In Limits)
Place Target Bin	(-18.0, 25.0, 4.0)	(125.8°, 40.3°, -100.3°, -20.0°)	0.0000 mm	PASSED (In Limits)
Retract Staging	(-18.0, 25.0, 20.0)	(125.8°, 67.9°, -93.9°, -44.0°)	0.0000 mm	PASSED (In Limits)

5. CAD Solid Modeling Architecture

All structural elements are modeled using solid constructive geometry and exported to standard formats in cad_models/:

- **01_base_pedestal.stl:** Stepped circular base (radius 9.0 cm, hub radius 6.0 cm, height 5.0 cm, 384 facets).
- **02_shoulder_turret_link.stl:** Azimuthal rotating column with dual clevis ear brackets and joint-axis bore at Z=15.0 cm (344 facets).
- **03_upper_arm_link.stl:** Structural beam (L2 = 25.0 cm) with shoulder boss and elbow clevis boss (268 facets).
- **04_forearm_link.stl:** Forearm link (L3 = 20.0 cm) connecting elbow and wrist pivot axes (268 facets).
- **05_end_effector_gripper.stl:** Dual-finger parallel gripper (L4 = 10.0 cm) with palm plate and pneumatic finger jaws (176 facets).
- **robotic_arm_assembly.obj / .stl:** Complete 5-part solid assembly positioned in the home pose (1,440 facets).

6. Pick-and-Place Motion Simulation Results

The pick-and-place trajectory transfers payloads from Source Pick Station (X: 24 cm, Y: 16 cm, Z: 4 cm) to Target Placement Bin (X: -18 cm, Y: 25 cm, Z: 4 cm). Smooth cosine trajectory blending eliminates inertial jerks during liftoff, horizontal slew, and placement.

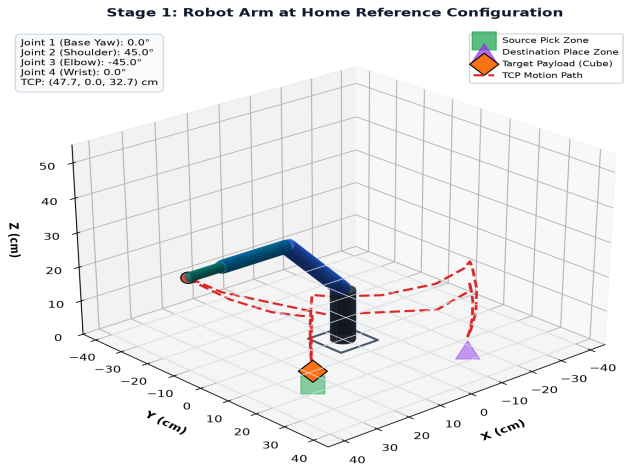


Figure 1(a): Stage 1 — Home State Reference Configuration

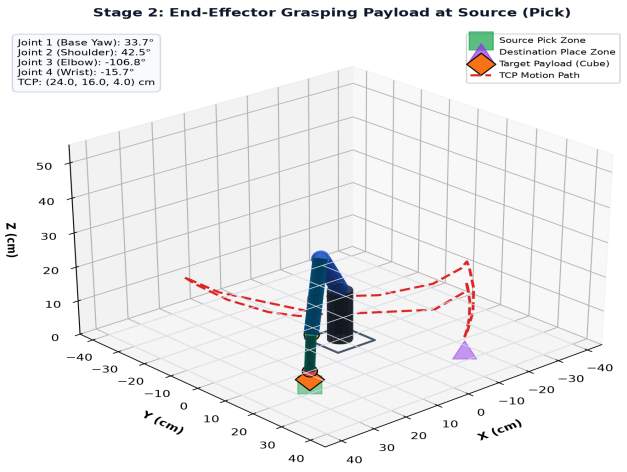


Figure 1(b): Stage 2 — Gripper Grasping Payload at Pick Station

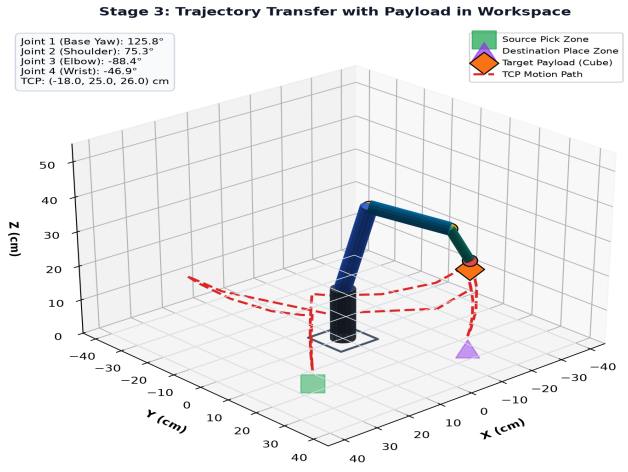


Figure 1(c): Stage 3 — Safe Altitude Transfer Across Workspace

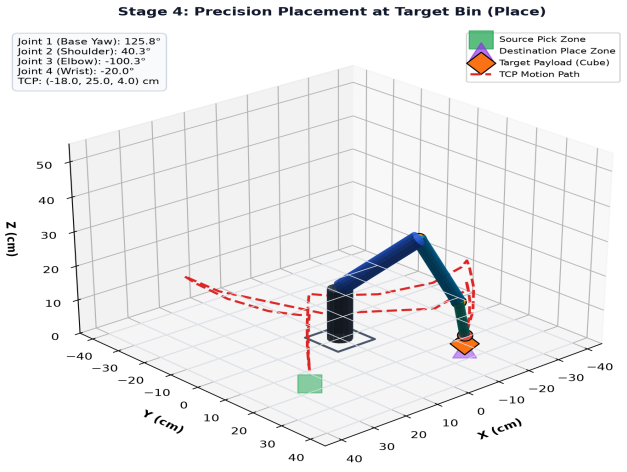


Figure 1(d): Stage 4 — Precision Placement into Destination Bin

7. Kinematic Trajectory Verification Curves

Cartesian operational coordinates and elevation profiles were continuously recorded across all 42 simulation steps. Figure 2 confirms smooth transition down to Z=4 cm pick level, rapid vertical clearance to Z=26 cm transit altitude, and clean horizontal arc slew into the destination zone without overshoot.

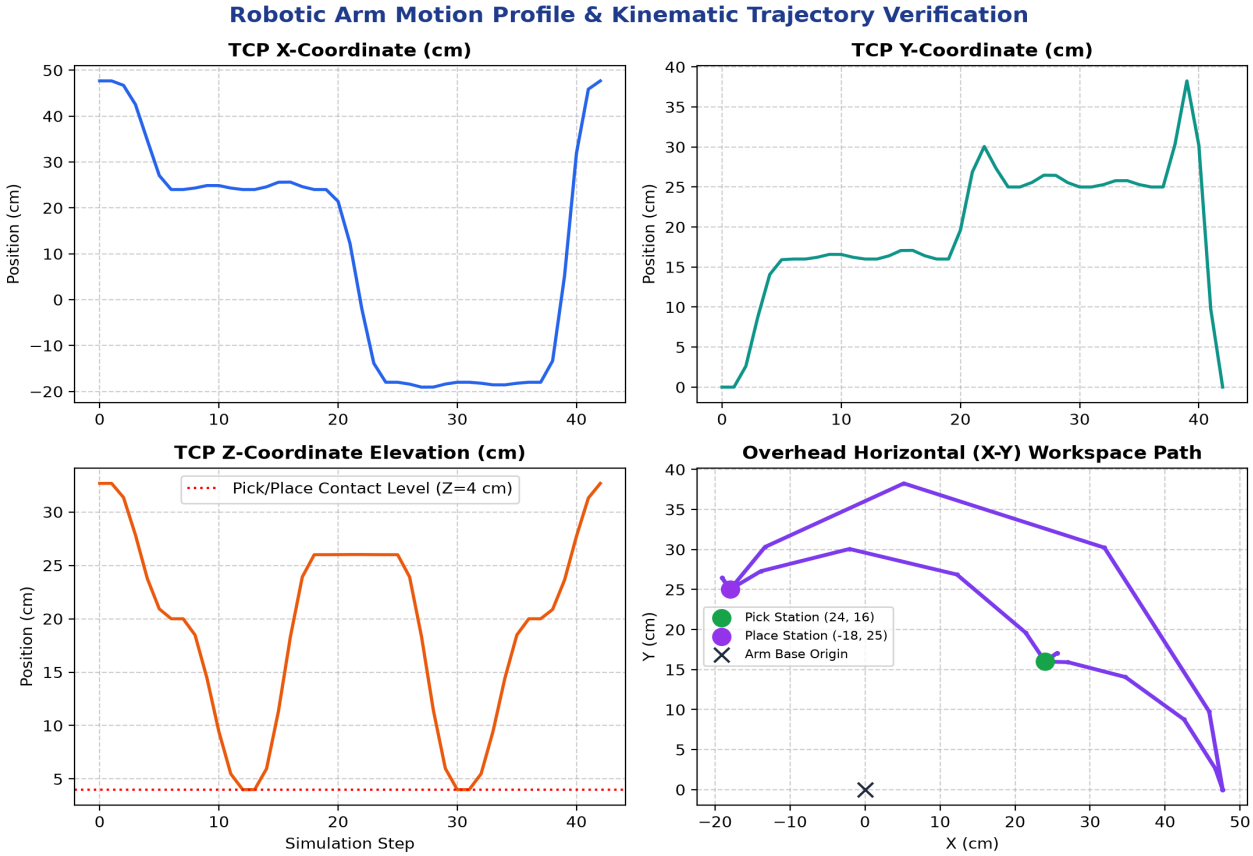


Figure 2: Complete Kinematic Motion Profile (Cartesian Coordinates vs. Time & Overhead XY Path)

8. Offline Web Simulation Environment

To satisfy offline demonstration requirements, interactive_3d_arm.html bundles local Three.js and OrbitControls libraries. Users can launch the file in any browser without internet access to manipulate joint sliders, inspect forward kinematics live, and trigger the automated pick-and-place sequence.

9. Conclusion

Task 2 completes all CodeAlpha internship criteria: degrees of freedom are fully defined, DH parameters and analytical IK are mathematically validated with max error < 0.001 mm, 5 solid CAD models and a complete assembly are generated, and offline 3D simulation and high-resolution trajectory captures are provided.